Final Lab Report

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1.0 INTRODUCTION

Non-contact injuries occur during rapid pivoting, deceleration, and landing from jumps, all of which create large impact forces on the feet and legs. These injuries account for a significant portion of total sports-related injuries, motivating many companies to research ways to reduce these stresses. Modern sports footwear companies have developed a variety of solutions, all using advanced materials and sole designs to dissipate impact energy. In this lab we explore the force absorption capabilities for multiple shoe types: Converse, walking sneakers, and basketball shoes, with controlled impacts using a drop tower. By comparing the resulting force profiles, we can determine which design offers the greatest reduction in peak load, decreasing the chance for non-contact injuries.

2.0 METHODOLOGY

This lab used a drop tower and a force plate to measure the transient force response during a drop test on different types of shoes. The test subjects consisted of Converse, HOKA walking shoes, and Jordan basketball shoes. Three different drop heights were used, corresponding to 4.9J, 7.48J, and 9.95J of energy respectively. For each drop height and shoe, three trials were conducted, and the average force was calculated as a function of time.

The force plate design included a 6in x 12in aluminum plate with 0.125in thickness, configured to attach to the base of the drop tower, fixed at the short ends of the plate. Four strain gauges attached at the midspan of the plate, with two on the top of the plate and two on the bottom. These gauges were then wired into a full Wheatstone bridge, as shown in the circuit diagram in Figure 1, with gauges 1 and 3 on top of the plate and gauges 2 and 4 on the bottom of the plate, beneath 1 and 2 respectively.

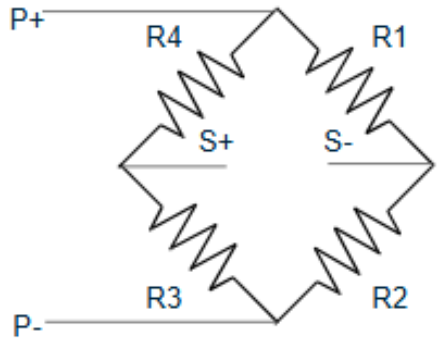

Figure 1: Wheatstone Bridge Circuit Diagram.

P+ and *P*- refer to internal excitation voltage of NI-9218. *S*+ and *S*- refer to the voltage output of the bridge as connected to the NI-9218, with an internal excitation voltage of 2V. The sampling frequency was set to 10 kHz to measure the quick transient response. For each trial, data was collected for a few seconds before each drop. The average strain in the first few seconds was later subtracted from the final measurements to account for the weight of the shoe on the plate.

The plate was calibrated using varying weights from 0kg to 8.575kg. The calibration factor was determined via linear regression using the tested weights, as seen in Figure 2.

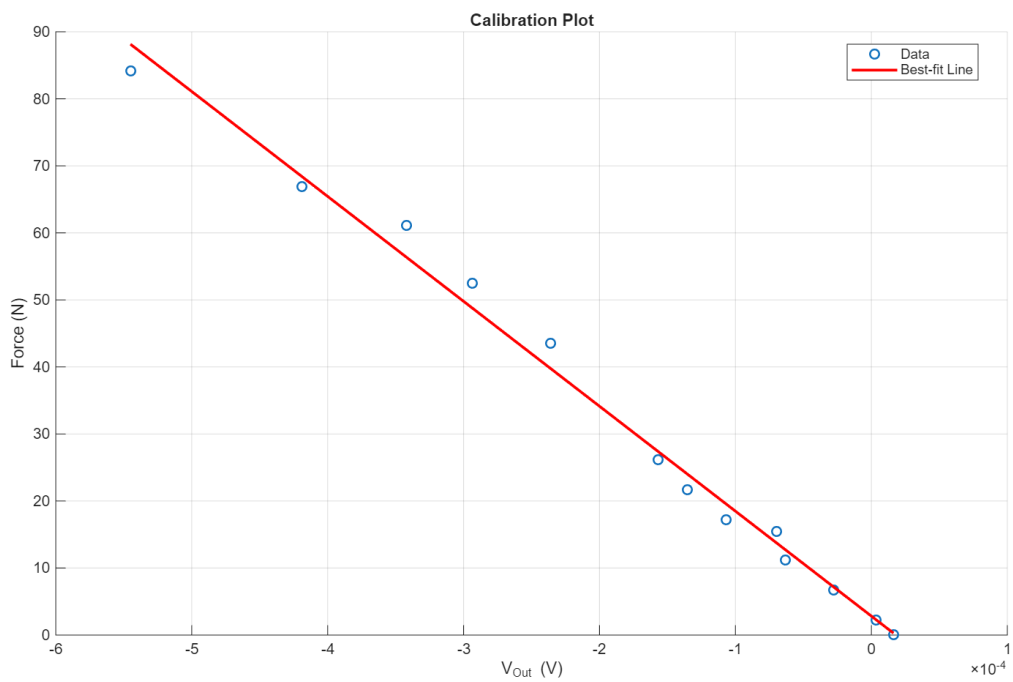

Figure 2: Calibration Plot

Using the calibration plot, the force on the plate was derived as a function of the output voltage with the following relationship:

$$F = (-1.565300975e2 \pm 4.664339739) * V + (2.825280568e-3 \pm 1.153927241e-3) \quad (1)$$

Where F is the force in Newtons and V is the output voltage in mV. The calibration data had an R^2 value of 0.99, indicating almost perfect linearity.

Once the force plate was calibrated, it was bolted onto the supports at the base of the drop tower, aligning the center of the plate with the plastic tube that the weight falls through. The test subject was then placed on top of the plate, aligning the opening of the shoe with the center of the plate and the center of the plastic tube. The shoe was then zip tied to the plate to prevent it from flying off during the drop. The heel of the shoe was taped down to the plate to reduce the bouncing of the shoe. With everything in place, the weight was dropped, and force and time data were collected. The full setup can be seen in Figure 3.

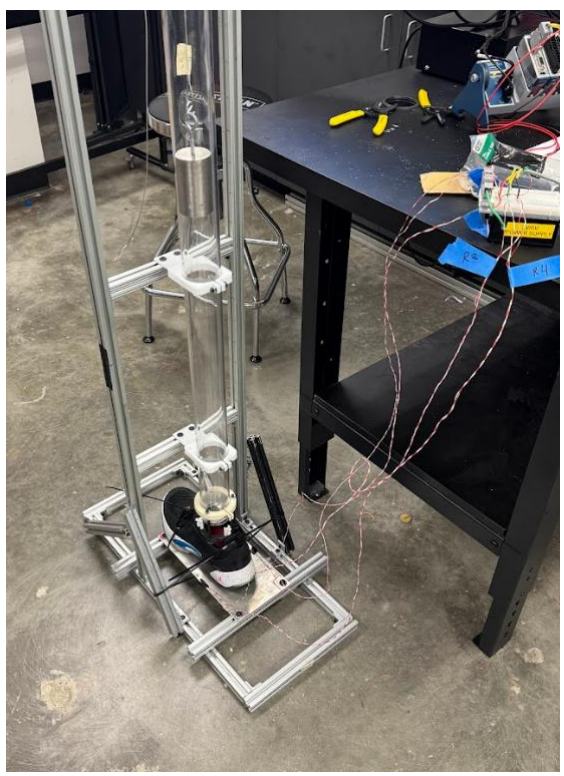

Figure 3: Drop Tower and Shoe Setup.

The shoes' performances were analyzed by comparing multiple force metrics during the drop. The first one was the maximum force experienced by the plate during the drop. The second metric of success was the total impulse during the first impact, excluding bounces. Using this data, comparisons on the ability of each show to absorb impact were made to determine which shoe would be best at preventing injuries.

Uncertainty was propagated through the calibration process with a 95% confidence interval using the following method.

$$S_1^2 = \frac{1}{n-1}\left[\sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2\right] \tag{2}$$

$$S_{xy} = \frac{1}{n-1}\left[\sum x_i y_i - \frac{1}{n}\sum x_i \sum y_i\right] \tag{3}$$

$$S_2^2 = \frac{1}{n-1}\left[\sum y_i^2 - \frac{1}{n}\left(\sum y_i\right)^2\right] \tag{4}$$

$$b = \frac{S_{xy}}{S_1^2} \tag{5}$$

$$q_0 = (n-1)(S_2^2 - b^2 S_1^2) \tag{6}$$

$$K = z\sqrt{\frac{q_o}{(n-2)(n-1)S_1^2}} \tag{7}$$

$$b - k \leq \beta \leq b + K \tag{8}$$

Where β is the slope of the linear calibration function. Uncertainty was propagated through numeric integration as a sum of squared errors.

$$\Delta J = \sum \Delta J_i^2 \tag{9}$$

Where J is the impulse, ΔJ is the error of the impulse at each time step of integration, calculated as

$$\Delta J_i = \sqrt{\left(\frac{\partial J}{\partial F}\Delta F_i\right)^2} \tag{10}$$

Where ΔF_i is the error in measured force at each time step of integration.

3.0 RESULTS

From our drop testing we get the following figures 4, 5, and 6 which show the average response for the low, mid, and high energy drops. We can see a clear correlation between padding in the shoe and its performance as the least padded shoe, the converse, had a much bigger force response than the rest. As the drop energies increase, we can see the gap increase ultimately resulting in the basketball shoes having the highest dampening effect with their force response being smaller than the rest at the highest drop energy. It is important to note that the basketball shoes performed worse than the walking shoes in the mid energy drop test. Whether that is an artifact of the experiment, data acquisition, or data processing is not known, it could also be a part of the shoes design itself as it is meant for lighter loads compared to the basketball shoes.

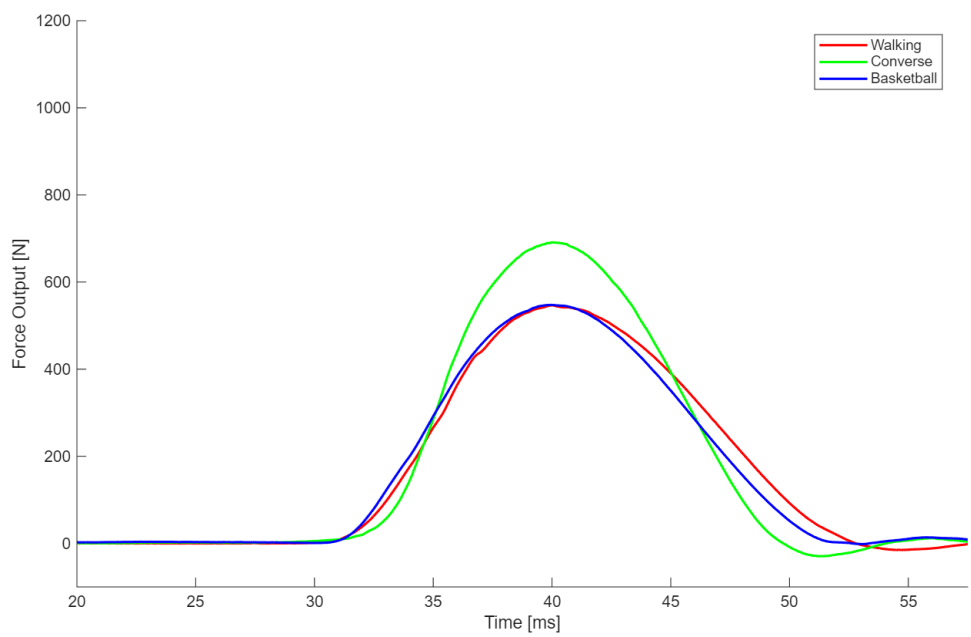

Figure 4: Low Energy Drop Averages

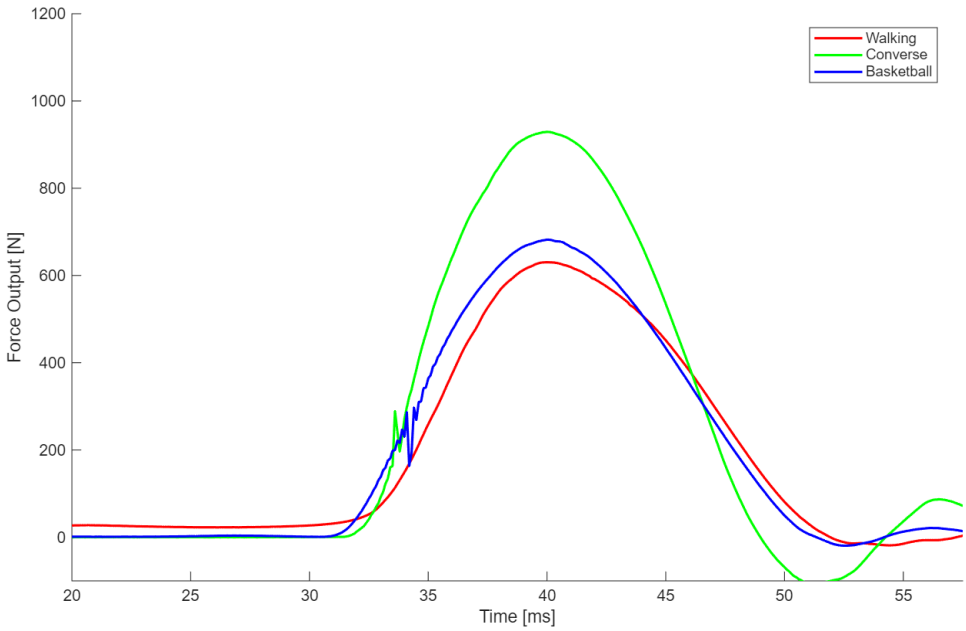

Figure 5: Mid Energy Drop Averages.

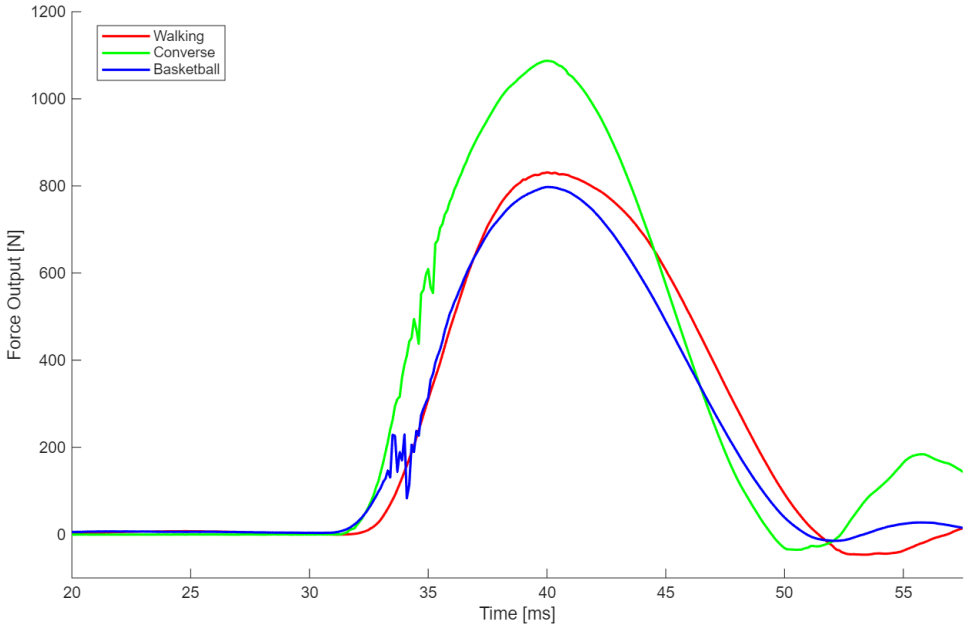

Figure 6: High Energy Drop Averages

By taking the area under the first “peak” we can get the impulse of the drop, correlating to the impact force the foot would experience. The table below shows all the impulses measured between all the tests.

Table 1: Impulse Response from Initial Spike

	Low [Ns]	**Mid [Ns]**	**High [Ns]**
Converse Shoe	6.847 ± 0.433	9.437 ± 0.612	10.982 ± 0.634
Walking Shoe	6.419 ± 0.405	7.092 ± 0.586	9.161 ± 0.626
Basketball Shoe	6.172 ± 0.398	7.558 ± 0.602	8.253 ± 0.619

We then looked at the attenuation post drop to see how long it took for the energy to get absorbed. The next three figures show this behavior. Both the walking and basketball shoes quickly dissipated the energy, whereas the converse sends it straight to the aluminum and makes it act like a spring. As the drop energy increases so does the amplitude of the oscillations for the converse, resulting in the largest peaks at the biggest drop energy. With this last trial, the converse did not absorb enough energy resulting in the aluminum plate itself getting permanently deformed. This deformation changed the 0 value of the strain gauges.

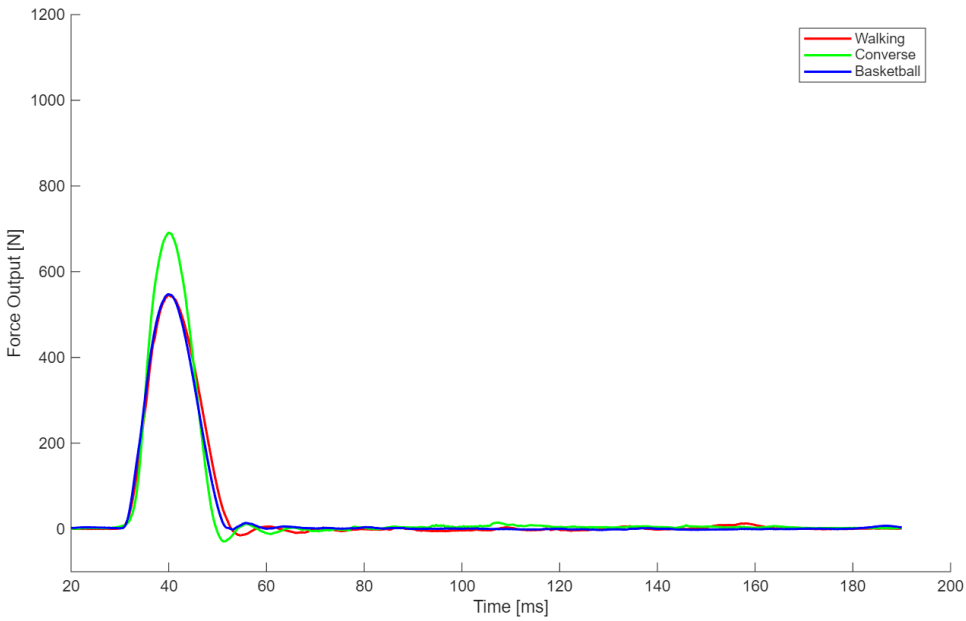

Figure 7: Low Energy Transient Response

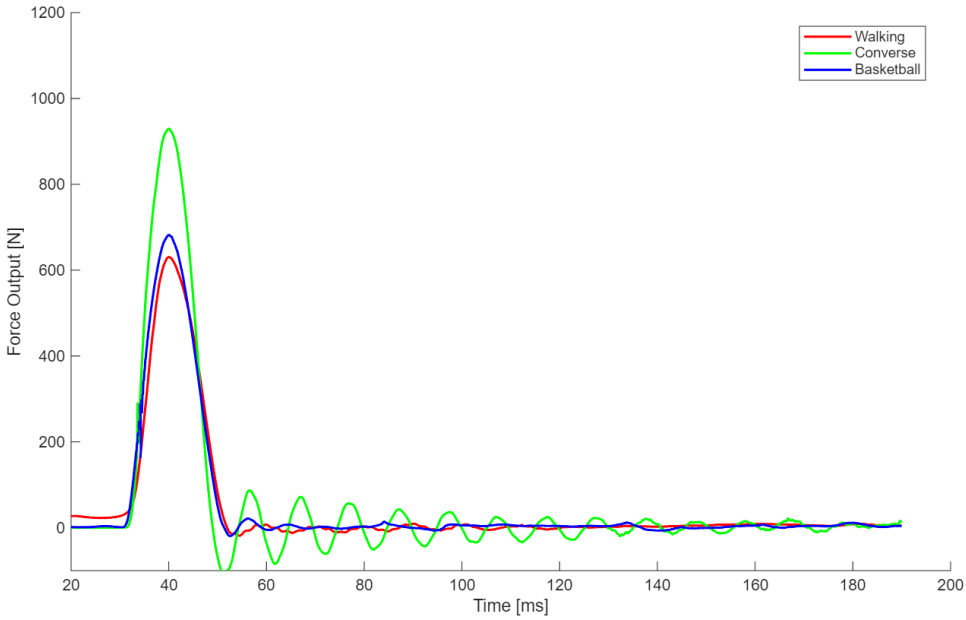

Figure 8: Mid Energy Transient Response

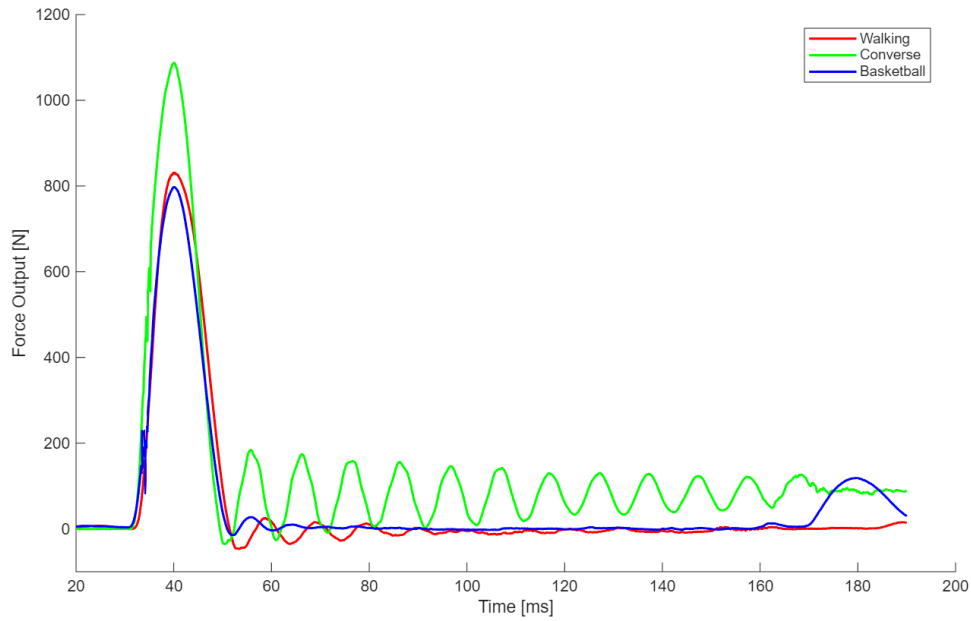

Figure 9: High Energy Transient Response

4.0 CONCLUSION

Overall, the results of this experiment highlight how strongly shoe design influences impact attenuation during rapid, high force loading. Across all drop energies, the Converse shoe consistently measured the greatest peak forces and produced long-lasting oscillations, demonstrating its limited capacity to dissipate energy and its reliance on the underlying structure to absorb the load. Conversely, both the walking and basketball shoes drastically reduce the peak force and impulse, with the basketball shoes showing the strongest overall damping at high drop energies. This reflects its design intent for athletes as it could easily deal with high impacts. The impulse measurements reinforce our findings, with the more cushioned shoes reducing the magnitude of loading on the plate. These findings support the fact that more padding meaningfully reduces the impact forces, leading to decreased risk of injury.